# When Retrieval Quality Decouples from Faithfulness: A Pre-Registered Audit of RAG Evaluation

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**Abstract**

Retrieval-augmented generation (RAG) papers often treat better retrieval quality as evidence that answers are more faithful. We show that the faithfulness scorer can be the under-identified variable: on 7,500 fixed generations, one intervention has effect sizes 0.011, 0.032, and 0.140 under DeBERTa, a second NLI scorer, and a local RAGAS-style judge. A 99-example two-rater human calibration reaches Cohen's $\kappa = 0.774$, but alignment with adjudicated support is weak for DeBERTa and moderate for the other scorers (Spearman $\rho = 0.140, 0.380, 0.441$); bootstrap intervals overlap, so the ordering is suggestive. We then use a pre-registered refinement-paradox stress test to audit identifiability once retrieval similarity, context-set structure, evaluator choice, human calibration, threshold transfer, and cost are separated. At $n = 500$ across five seeds, the DeBERTa-measured SQuAD/Mistral pilot effect shrinks to 1.1 faithfulness points, with only 1/5 seeds significant. A matched HIGH/LOW context-coherence test does not support a causal CCS interpretation (HIGH–LOW $= -0.002$, $p = 0.628$, 95% CI $[-0.022, 0.017]$); HIGH-CCS contexts hallucinate more often (16.5% vs. 9.0%). Threshold transfer is uneven, and HCPC-v2, CRAG, and RAPTOR-2L yield no single deployment ordering once tail latency and indexing cost are reported; Self-RAG is documented separately as a harness-mismatched appendix baseline. Coherent uninformative noise hurts less than random off-topic noise. We propose a minimum reporting standard for RAG faithfulness claims (paired query-level contrasts, retrieval/context statistics, multiple scorers, human calibration on disagreements, threshold-transfer evidence, hallucination rates, p99 latency, indexing cost) and release ControlledRAG as the audit protocol that operationalizes it.

## 1 Introduction

RAG evaluation has a familiar shortcut: a retrieval score improves, one automatic faithfulness metric moves in the desired direction, and the paper infers that answer support has improved. Our audit finds that the scorer can be the fragile link. On the same 7,500 generated answers, the same intervention has effect sizes 0.011, 0.032, and 0.140 under DeBERTa, a second NLI scorer, and a local RAGAS-style judge. In a 99-example two-rater calibration ($\kappa = 0.774$), DeBERTa has weak rank alignment with adjudicated support labels, while the second NLI scorer and RAGAS-style judge are only moderately aligned; their confidence intervals overlap, so the ordering is suggestive rather than definitive. The durable finding is not that one scorer is best, but that RAG faithfulness claims are not identified by a single observer.

This paper audits that gap. A generator receives a context set, not a list of independent passage scores, and a faithfulness conclusion depends on the observer used to score support. We use the refinement paradox as a stress test: the pilot finding that more aggressive per-passage refinement can raise retrieval similarity while lowering measured answer faithfulness. The goal is not to promote that phenomenon as large or stable. It is to ask a sharper methodological question:

> *Which RAG faithfulness claims remain identifiable once local retrieval similarity, context-set structure, evaluator choice, human calibration, threshold transfer, and cost are separated?*

The answer is sobering. The strongest intuitive story – that context coherence explains faithfulness failures – fails under a direct control. We construct HIGH- and LOW-CCS context sets matched on mean query-passage similarity; the coherence manipulation is large, but faithfulness does not improve. This failure is the point. It shows how a plausible retrieval-time quantity can look meaningful in aggregate tables and then lose its apparent effect once the confounder is isolated.

The wider audit reaches the same conclusion from other angles. The pilot DeBERTa effect becomes small and seed-sensitive at $n = 500 \times 5$. The same generations imply different intervention magnitudes under different scorers. A small two-rater human calibration clarifies that the scorer disagreement is not cosmetic: no automatic scorer has strong human alignment in this sample, although the RAGAS-style judge has the highest rank correlation. Thresholds tuned on one dataset transfer unevenly. Baseline comparisons change when p99 latency and indexing cost are included.

**Recommended minimum reporting standard for RAG faithfulness claims.**

1. Paired query-level contrasts with bootstrap intervals.
2. Local query-passage similarity and at least one set-level context statistic.
3. Multiple faithfulness scorers.
4. A small human calibration whenever automated metrics disagree.
5. Threshold transfer across datasets.
6. Hallucination rates with Wilson intervals.
7. Mean latency, p99 latency, and offline indexing cost.

**Contributions.**

- We show that faithfulness scoring is metric-fragile on fixed generations, and calibrate the disagreement with a 99-example two-rater human audit.
- We provide a pre-registered scale audit of a pilot refinement-paradox cell, replacing a small-sample headline with $n = 500 \times 5$ seed evidence.
- We falsify the strong CCS interpretation with a matched-similarity intervention, while preserving a narrower diagnostic role via a coherent-versus-random noise test.
- We audit threshold transfer and cost-aware baselines, and document the Self-RAG artifact separately as a harness-mismatched appendix baseline.
- We package the checks as ControlledRAG, a reusable protocol for auditing RAG faithfulness claims. Code and data are anonymized in the supplementary material and will be released upon acceptance.

The paper's thesis is deliberately conservative: RAG faithfulness claims are under-identified unless retrieval similarity, context-set structure, evaluator choice, human calibration, threshold transfer, and cost are reported separately.

# 2 Background and Problem Setup

**RAG faithfulness.** RAG systems retrieve passages and condition a generator on the resulting context [Lewis et al., 2020, Karpukhin et al., 2020]. A generated answer is faithful when its substantive claims are supported by the retrieved context, not merely when the answer is true in the world. This distinction matters because a model can answer from parametric memory even when retrieval failed, or can produce unsupported details despite apparently relevant passages [Maynez et al., 2020, Ji et al., 2023].

**Local retrieval similarity.** Most retrieval metrics are local: they score each query-passage pair with embedding similarity, cross-encoder relevance, or reranking rank. Local scores are useful, but they do not identify answer support. Three individually relevant passages may be redundant, contradictory, or missing a bridge fact. One lower-scoring passage may contain the answer.

**Context coherence and CCS.** The Context Coherence Score (CCS) is a retrieval-time statistic over a retrieved context set $C$: $\mathrm{CCS}(C) = \mathrm{mean}_{i<j} \cos(e_i, e_j) - \mathrm{sd}_{i<j} \cos(e_i, e_j)$, where $e_i$ and $e_j$ are passage embeddings. The mean term rewards mutual topical compatibility, while the standard-deviation penalty reduces scores for uneven sets with one coherent sub-cluster and one outlying passage. CCS asks a set-level question that local relevance does not: do the passages look mutually compatible? The audit treats CCS as a diagnostic candidate. It does not assume that CCS causes faithfulness, and the matched test in Section 4 rejects that strong interpretation.

**Identifiability.** We use "identified" in a practical evaluation sense. A RAG claim is not identified if another unreported factor could plausibly explain the reported effect. A local retrieval improvement does not identify support if context-set structure is uncontrolled. A single automatic metric does not identify an effect size if other faithfulness observers disagree. A tuned threshold does not identify a reusable policy if off-diagonal transfer is poor. A method does not identify deployment value if p99 latency and indexing cost are omitted.

**Why the usual evidence is risky.** The risky combination is small samples, one scorer, threshold tuning on the reporting domain, no human calibration, and partial cost reporting. Each choice can be reasonable in isolation. Together, they let a paper report a clean faithfulness story without separating the variables needed to support it.

# 3 Pre-Registered Audit Protocol: ControlledRAG

The experiments are a controlled falsification suite. Each test isolates a claim that can look convincing in a conventional aggregate RAG table, then asks whether the claim survives once the relevant confounder is separated.

**Pre-registration and statistics.** Sample sizes, seeds, primary statistics, and decision rules were frozen before execution. Paired continuous contrasts use Wilcoxon signed-rank tests and Cohen's $d_z$. Means use 10,000-resample bootstrap confidence intervals. Hallucination rates use Wilson intervals. Thresholds are frozen except in the threshold-transfer experiment, where tuning and evaluation datasets are explicitly separated. No paid API calls are used in the revision results.

**Datasets, generators, and scorers.** The core revision evidence uses short-answer extractive QA, primarily SQuAD, PubMedQA, HotpotQA, NaturalQuestions, and TriviaQA [Rajpurkar et al., 2016, Jin et al., 2019, Yang et al., 2018, Kwiatkowski et al., 2019, Joshi et al., 2017]. The main scaled cell is SQuAD with Mistral-7B-Instruct [Jiang et al., 2023]. Faithfulness is scored with DeBERTa-v3 NLI [He et al., 2021], a second NLI model, and a local RAGAS-style Mistral-as-judge rubric [Es et al., 2024]. The local judge is useful for triangulation but may share family-level biases with the generator; we treat that as a limitation rather than a source of authority.

**Human calibration design.** To ground the scorer disagreement, two raters annotate 99 examples sampled from the fixed-generation multi-metric cell. Labels are `supported`, `partially_supported`, and `unsupported`; adjudicated labels are mapped to $1, 0.5, 0$ for rank correlations. The instructions require raters to judge support using only the provided passages.

Table 1: ControlledRAG audit controls. Each control can be reused with other datasets, retrievers, generators, and faithfulness metrics.

| Control | Claim it tests | Failure mode exposed | Primary evidence |
|---|---|---|---|
| Similarity-matched intervention | CCS improves faithfulness after local relevance is fixed | A set statistic can lose its apparent effect once mean query-passage similarity is controlled | Table 2 |
| Multi-seed scale correction | A pilot effect remains stable under repeated sampling | Small effects can be seed-sensitive even when the pilot direction is plausible | Table 3 |
| Fixed-generation multi-metric and human-calibrated scoring | One automatic judge identifies the intervention effect size | The same outputs can imply different magnitudes, and human alignment can be only moderate | Tables 4–6 |
| Threshold-transfer matrix | A tuned threshold is a reusable policy | In-domain tuning can hide weak off-diagonal transfer | Table 7 |
| Coherent-vs-random noise separation | Coherence is just generic retrieval quality | Same-topic uninformative context and random context can have different slopes | Table 8 |
| Cost-aware baseline audit | A method is deployment-preferable | Dataset wins can disappear once hallucination, p99 latency, and indexing cost are included | Figure 2; Table 9 |

# 4 Matched-Similarity Causal Test

The strong CCS hypothesis predicts that, at fixed local retrieval similarity, higher context coherence should improve faithfulness. We test that directly. For each SQuAD query, the protocol retrieves candidate passages and chooses a HIGH-CCS and LOW-CCS three-passage set whose mean query-passage similarity differs by at most 0.02. The construction succeeds: mean absolute similarity gap is 0.006 and mean CCS gap is 0.533 across 200 matched pairs.

Table 2: Matched-similarity intervention. Higher CCS does not improve faithfulness at fixed mean query-passage similarity.

| Setting | $n$ | HIGH faith | LOW faith | HIGH–LOW | $p$ | 95% CI |
|---|---|---|---|---|---|---|
| SQuAD / Mistral / DeBERTa | 200 | 0.636 | 0.639 | -0.002 | 0.628 | [-0.022, 0.017] |

The binary hallucination result points in the wrong direction for the strong claim: HIGH-CCS contexts hallucinate more often than LOW-CCS contexts (16.5% vs. 9.0%). We therefore reject the causal CCS interpretation tested here. CCS may still be diagnostically useful, but it is not sufficient to identify answer support once local retrieval similarity is held fixed.

A span-presence check suggests a simple reason for the wrong-sign result. We normalize the gold answer and ask whether it appears anywhere in the three retrieved passages. LOW-CCS contexts contain the answer more often than HIGH-CCS contexts (27.0% vs. 17.5%; exact McNemar $p = 0.011$); the four-way count is 19 pairs with the answer in both, 35 in LOW only, 16 in HIGH only, and 130 in neither.

Thus the HIGH-CCS sets do not merely provide an equally answer-bearing but more coherent scaffold; they more often omit the answer span. A plausible reading is that coherent-looking context can still be incomplete, encouraging fluent unsupported answers when the needed span is absent. We report this as a hypothesis suggested by the data, not as a tested mechanism.

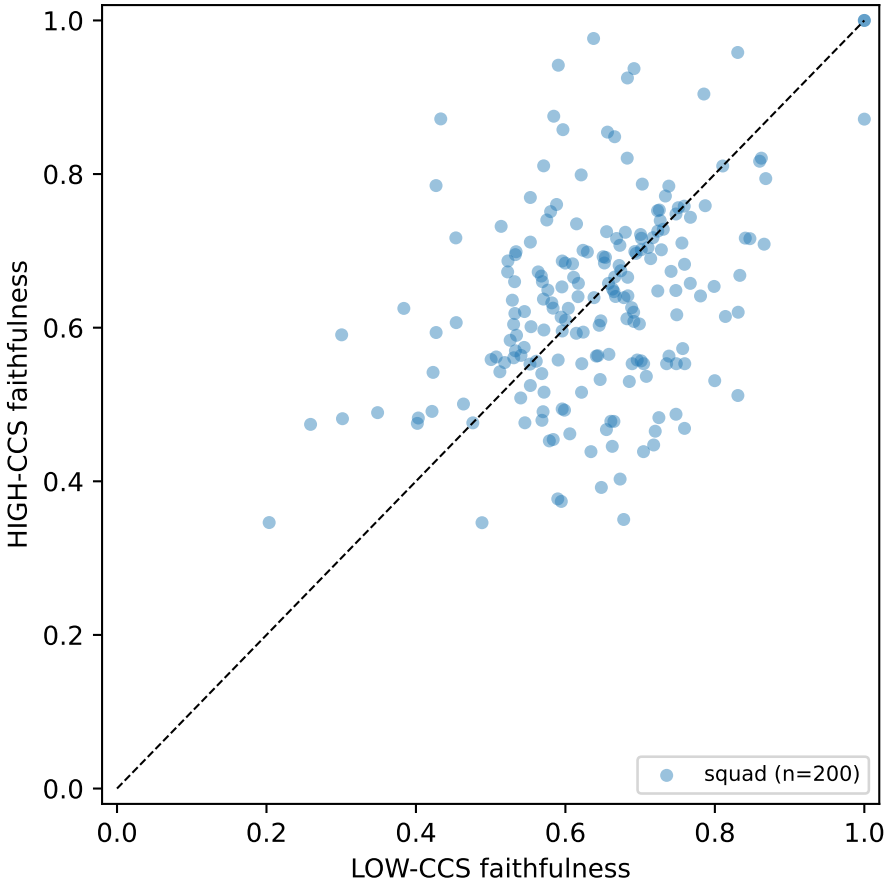


Figure 1: Paired HIGH–LOW CCS faithfulness differences under the matched-similarity intervention. The diagonal line represents equal HIGH and LOW faithfulness; points above the line are pairs where HIGH-CCS helped, and points below it are pairs where it hurt. The distribution straddles the diagonal, so the controlled test rejects the strong CCS hypothesis rather than supporting it.

## 5 Scaling the Pilot Paradox

The pilot refinement paradox is interesting only if it survives adequate sampling. We therefore rerun the SQuAD/Mistral cell at $n = 500$ over five seeds and three conditions: dense baseline, HCPC-v1 refinement, and HCPC-v2 gated refinement. Under the DeBERTa scorer, the baseline-minus-HCPC-v1 effect is 1.1 faithfulness points, and only one of five seeds reaches $p < 0.05$ for the paired contrast.

Table 3: Scaled SQuAD/Mistral headline cell. The pilot direction survives as a small DeBERTa-measured non-monotonicity, not as a large stable effect.

| Condition | $n$ | Faithfulness | Hallucination | Retrieval similarity |
|---|---|---|---|---|
| Baseline | 2500 | 0.661 | 0.146 | 0.532 |
| HCPC-v1 | 2500 | 0.650 | 0.147 | 0.569 |
| HCPC-v2 | 2500 | 0.661 | 0.143 | 0.532 |

The result should be read as a scale correction. HCPC-v1 raises retrieval similarity while lowering DeBERTa faithfulness, but the magnitude is small and seed-sensitive. HCPC-v2 recovers the HCPC-v1 drop on average, yet this does not establish a general solution to hallucination or a global method ranking.

# 6 Faithfulness Scoring Is Metric-Fragile

The same generated answers can tell different stories depending on the faithfulness observer. We therefore rescore the full 7,500-row scaled cell with DeBERTa-v3, a second NLI model, and a local RAGAS-style judge while holding the generations fixed.

For correlation uncertainty, we use 10,000 paired-index bootstrap resamples with a fixed seed and percentile 95% intervals. Each resample preserves the pairing between the two scores being correlated; human-label correlations resample the adjudicated ordinal labels and automatic score together.

Table 4: Identical generations produce different intervention magnitudes under different faithfulness judges.

| Condition | $n$ | DeBERTa | Second NLI | RAGAS-style judge |
|---|---|---|---|---|
| Baseline | 2500 | 0.661 | 0.350 | 0.730 |
| HCPC-v1 | 2500 | 0.650 | 0.318 | 0.590 |
| HCPC-v2 | 2500 | 0.661 | 0.351 | 0.728 |
| Baseline–v1 | – | 0.011 | 0.032 | 0.140 |

Table 5: Metric agreement on the same 7,500 generations. Brackets give paired bootstrap 95% CIs.

| Metric pair | Pearson $r$ [95% CI] | Spearman $\rho$ [95% CI] |
|---|---|---|
| DeBERTa / second NLI | 0.259 [0.240, 0.278] | 0.265 [0.245, 0.285] |
| DeBERTa / RAGAS-style judge | 0.182 [0.162, 0.202] | 0.212 [0.191, 0.233] |
| Second NLI / RAGAS-style judge | 0.674 [0.661, 0.686] | 0.651 [0.638, 0.665] |

This is the central positive result. The DeBERTa effect is small; the second NLI effect is larger; the RAGAS-style effect is much larger. Because DeBERTa and the RAGAS-style judge correlate weakly on the same examples, the issue is not just a monotone rescaling. Evaluator choice changes the empirical conclusion.

## 6.1 Human Calibration of Metric Disagreement

To calibrate the scorer disagreement, two raters annotate 99 examples from the fixed-generation cell. The raw agreement is 0.919 and Cohen's $\kappa$ is 0.774. Adjudicated labels are 75 supported, 20 partially supported, and 4 unsupported. Spearman correlation with the adjudicated ordinal labels is weak for DeBERTa, moderate for the second NLI scorer, and moderate for the RAGAS-style judge.

Table 6: Human calibration audit ($n = 99$, two raters, adjudicated labels). Brackets give paired bootstrap 95% CIs.

| Metric | Spearman $\rho$ [95% CI] | Pearson $r$ [95% CI] | Notes |
|---|---|---|---|
| DeBERTa-v3 NLI | 0.140 [-0.047, 0.314] | 0.145 [-0.016, 0.296] | Weak alignment |
| Second NLI | 0.380 [0.208, 0.534] | 0.385 [0.225, 0.523] | Moderate alignment |
| RAGAS-style judge | 0.441 [0.249, 0.624] | 0.338 [0.121, 0.542] | Moderate alignment |

The human calibration does not settle faithfulness evaluation generally. It does rule out a comfortable interpretation in which any one automatic metric can be treated as obviously correct. In this sample, the RAGAS-style judge is closest to human rank order, but only moderately; DeBERTa is poorly aligned. The pairwise differences in rank alignment between scorers have overlapping bootstrap intervals, so the relative ordering should be read as suggestive rather than definitive. The safe conclusion is that metric disagreement must be reported, and small single-metric effects should not carry broad claims.

# 7 Threshold Transfer

Thresholded retrieval policies can look strong when the threshold is tuned and evaluated on the same dataset. We test transfer by tuning $\tau$ on each of five datasets and evaluating on all five. For evaluation dataset $d$ and

threshold $\tau$, recovery is

$$\text{Recovery}(d, \tau) = \frac{\text{Faith}_{\text{gate}}(d, \tau) - \text{Faith}_{\text{HCPC-v1}}(d)}{\text{Faith}_{\text{baseline}}(d) - \text{Faith}_{\text{HCPC-v1}}(d)}.$$

Scores can exceed 1 when the gated policy exceeds the baseline relative to HCPC-v1, and can be negative when transfer is worse than HCPC-v1 under this normalization.

Table 7: Threshold-transfer recovery score. SQuAD, PubMedQA, and NaturalQuestions show large in-domain versus off-diagonal gaps; TriviaQA and HotpotQA do not. A recovery score of 1.452 on PubMedQA (diagonal) means the gated policy exceeded the dense baseline by 45% of the baseline-minus-HCPC-v1 gap. A negative score means the gated policy underperforms HCPC-v1 under this normalization.

| Tune dataset | Diagonal recovery | Off-diagonal mean | Gap |
|---|---|---|---|
| PubMedQA | 1.452 | 0.451 | 1.002 |
| NaturalQuestions | 1.401 | 0.211 | 1.191 |
| SQuAD | 0.823 | -0.141 | 0.963 |
| TriviaQA | 0.464 | 0.698 | -0.234 |
| HotpotQA | -0.112 | 0.093 | -0.205 |

The result is uneven transfer, not a universal failure. Some datasets show a large in-domain advantage, while others do not. Reporting only the tuned diagonal would make the policy look more stable than the off-diagonal evidence supports.

# 8 What Survives: Coherence vs. Similarity

The matched-similarity test rejects the strong CCS interpretation, but it should not be overread as evidence that context-set structure is useless. We compare two kinds of uninformative context: random off-topic noise and same-topic passages that preserve some contextual coherence while omitting the answer. If coherence were only generic retrieval quality under another name, the slopes should look similar.

Table 8: Noise slope response. Coherence-preserving uninformative noise hurts less than random off-topic noise at matched rate.

| Condition | Faith slope | Similarity slope | Drop at full noise |
|---|---|---|---|
| Random noise | -0.069 | -0.481 | 0.052 |
| Coherent uninformative noise | -0.043 | -0.113 | 0.042 |

This supports a narrower claim. Coherence carries diagnostic signal beyond mean query similarity, but it does not identify support by itself. A coherent context set can still omit the answer, and a low-coherence set can still contain the necessary evidence.

# 9 Cost-Aware Baseline Audit

We compare HCPC-v2, CRAG [Yan et al., 2024], RAPTOR-2L [Sarthi et al., 2024] in a matched short-answer harness. Self-RAG [Asai et al., 2024] is reported separately in the supplement: its released fine-tuned Llama-2-7B checkpoint, 8-bit inference path, reflective decoding behavior, and entailment-style scoring make that comparison harness-conditional.

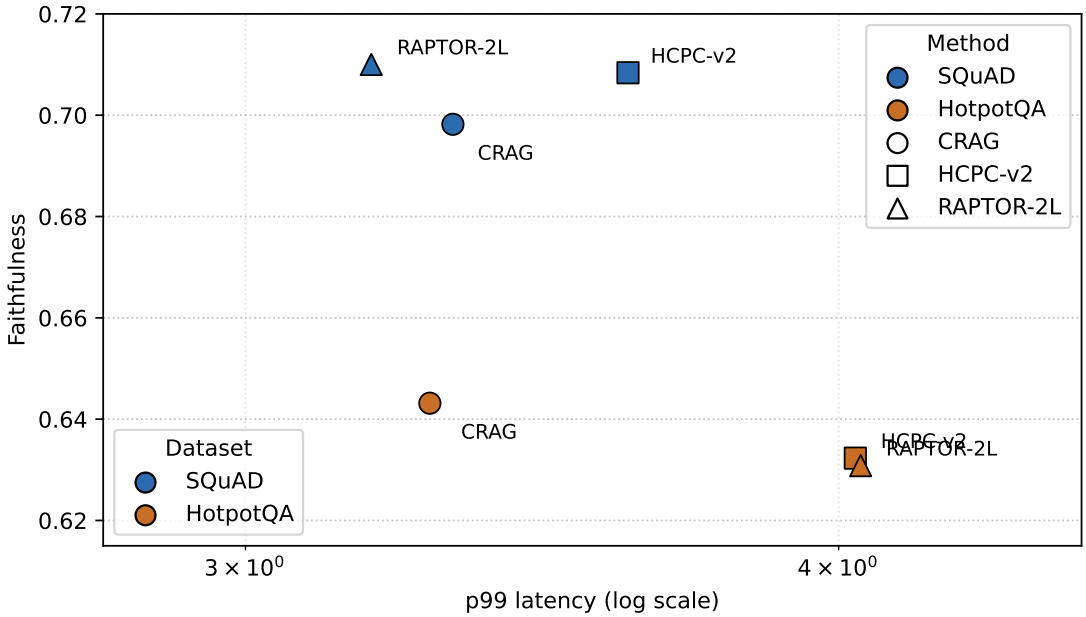


Figure 2: Cost-aware view of the full baseline audit. The x-axis is p99 latency on a log scale. No matched-harness method has a single deployment ordering across datasets once faithfulness, hallucination behavior, tail latency, and offline indexing costs are considered. The harness-mismatched Self-RAG numbers are reported separately in the supplement.

Table 9: Head-to-head baseline comparison, $n = 200$ per dataset/condition. Latency is milliseconds.

| Dataset | Method | Faith | Halluc. | Mean lat. | p99 lat. |
|---|---|---|---|---|---|
| SQuAD | CRAG | 0.698 | 0.120 | 1021 | 3318 |
| SQuAD | HCPC-v2 | 0.708 | 0.125 | 1349 | 3612 |
| SQuAD | RAPTOR-2L | 0.710 | 0.125 | 1296 | 3189 |
| HotpotQA | CRAG | 0.643 | 0.105 | 1473 | 3281 |
| HotpotQA | HCPC-v2 | 0.632 | 0.130 | 1841 | 4032 |
| HotpotQA | RAPTOR-2L | 0.631 | 0.130 | 1906 | 4043 |

HCPC-v2 is competitive on SQuAD, where RAPTOR-2L is slightly higher in faithfulness and lower in p99 latency. On HotpotQA, CRAG has higher faithfulness, lower hallucination, and lower latency. Self-RAG numbers are in the supplement; a fair evaluation requires a scoring protocol that accounts for reflective decoding rather than treating every extra token as ordinary answer content.

Table 10: RAPTOR cost audit across three datasets, $n = 100$ each.

| Dataset | Faith | Halluc. | p50 lat. | p99 lat. | Dense idx. | RAPTOR idx. |
|---|---|---|---|---|---|---|
| SQuAD | 0.789 | 0.050 | 1190 | 4943 | 8.333 s | 161.203 s |
| PubMedQA | 0.560 | 0.290 | 3901 | 6812 | 1.177 s | 108.264 s |
| HotpotQA | 0.617 | 0.210 | 1973 | 4274 | 7.140 s | 99.740 s |

## 10 Discussion

**The collapse is informative.** The audit weakens the simple refinement-paradox story: the pilot effect becomes small under DeBERTa, the matched CCS test rejects the strong coherence interpretation, and the baseline ranking changes with dataset and cost surface. That is the point. Standard RAG reporting under-identifies faithfulness claims.

**What can and cannot be claimed.** Local retrieval quality and answer faithfulness can decouple; CCS is diagnostic, not causal; metric choice materially changes intervention magnitude; human calibration gives only moderate alignment for the closest automatic scorer; threshold transfer is uneven; and baseline comparisons require hallucination, p99 latency, and indexing cost. The audit does not show that HCPC-v2 solves hallucination, that the DeBERTa paradox is large and stable at scale, that Self-RAG is globally weak, or that any automatic scorer is a universal faithfulness metric. The operational recommendation is the seven-item reporting standard in Section 1.

# 11 Related Work

RAG combines parametric generation with retrieved evidence [Lewis et al., 2020, Karpukhin et al., 2020]. Faithfulness evaluation often relies on NLI-style entailment or judge rubrics [He et al., 2021, Es et al., 2024], while CRAG, Self-RAG, and RAPTOR instantiate corrective retrieval, reflective generation, and hierarchical summary retrieval [Yan et al., 2024, Asai et al., 2024, Sarthi et al., 2024]. Our contribution is not a new retriever or a universal ranking; it is an audit protocol for separating retrieval similarity, context-set structure, scorer choice, human calibration, transfer, and cost.

# 12 Limitations and Ethics

**Short-answer scope.** The main evidence concerns short-answer extractive QA. It should not be used to claim the same behavior in long-form synthesis, agentic workflows, or domain-specific retrieval without additional controls.

**Human calibration and judge bias.** The human evaluation has 99 examples, two raters, and adjudicated labels: it calibrates scorer disagreement in the audited cell, but does not definitively validate any faithfulness metric. The RAGAS-style judge is local and zero-dollar, but may share family-level biases with the generator; the stronger conclusion is metric fragility, not that the judge is universally correct.

**CCS and threshold scope.** The matched-similarity test is a single-domain causal test on SQuAD/Mistral. It rejects the strong CCS interpretation in that setting; it does not exhaust all possible context-structure hypotheses. Threshold transfer is also uneven, so thresholded policies should report off-diagonal behavior.

**Self-RAG harness fairness.** Self-RAG is a harness-mismatched appendix baseline because it uses a different backbone, quantization path, and reflective decoding objective than the Mistral-backed methods in the main table. Its measured behavior should not be read as a global statement about Self-RAG.

**Large-model reproduction.** The pilot's 70B and 120B observations are not re-verified under the zero-dollar revision constraint and therefore are not treated as central evidence; they are not contradicted by the audit.

**Ethics.** The audit is intended to reduce overclaims about faithful RAG systems. The main risk is misuse: a single metric, a single threshold, or one favorable baseline row could be cherry-picked to market a system as grounded. We therefore emphasize per-query logs, multiple metrics, human calibration, negative results, and cost-aware reporting.

# 13 Conclusion

This paper turns a weakened refinement-paradox story into a stronger audit result. Better local retrieval quality, higher context coherence, a single faithfulness metric, an in-domain threshold, or one favorable baseline row does not identify RAG answer support. Under ControlledRAG, the safe conclusion is not that one method wins, but that RAG faithfulness claims need matched controls, scale checks, multi-metric and

human-calibrated scoring, transfer tests, and cost accounting before they are treated as evidence of grounded generation. The central durable contribution is the fixed-generation metric-fragility finding: the same outputs support materially different intervention magnitudes under different scorers, and human calibration shows that none can be treated as an unqualified ground truth for faithfulness.

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